SooHyun Yang

Google Scholar: https://scholar.google.com/citations?user=_zczJPkAAAAJ&hl=en
Github: https://github.com/Yang-SooHyun
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EDUCATION

University of Seoul **Seoul, South Korea**
Master of Science in Environmental Engineering Sep. 2025 – Feb. 2027
GPA: 4.4/4.5
Thesis: Construction of a long-term, high spatiotemporal resolution satellite-derived chlorophyll-*a* dataset for global inland lakes
Advisor: YoonKyung Cha

University of Seoul **Seoul, South Korea**
Bachelor of Science in Environmental Engineering Mar. 2020 – Aug. 2025
GPA: 3.8/4.5
Thesis: Development of Bayesian Network for Probabilistic Mid-to Long-term Forecasting for Harmful Algal Blooms

RESEARCH INTERESTS

Processing satellite observations, implementing robust quality control frameworks, and constructing long-term high-resolution geospatial datasets.
Analyzing long-term variability, identifying trends on the Earth's surface, and modeling spatiotemporal environmental dynamics under climate conditions.
Applying advanced neural network architectures for spatiotemporal data imputation, time-series reconstruction, and predictive environmental modeling.
Leveraging spatial deep learning, semantic segmentation, and object detection for automated feature extraction, land classification, and surface anomaly detection.

RESEARCH EXPERIENCE

University of Seoul **Seoul, South Korea**
Graduate Research Assistant, Water Environment Management Lab Sep. 2025 – Feb. 2027

- Developed a cyanobacteria cell counts forecasting model utilizing deep learning models (Transformer, TFT, TCN, LCA).
- Imputed daily water quality and cyanobacteria cell counts data from mixed-frequency inputs using SAITS (Self-Attention-based Imputation for Time Series).
- Developed a transformer-based hierarchical deep learning model for satellite Chlorophyll-*a* prediction.
- Calibrated and adapted established bio-optical algorithms to optimize Chlorophyll-*a* retrieval accuracy for the unique optical properties of diverse global lakes.
- Constructed GLCD250-MOD, a daily, 250 m Chlorophyll-*a* dataset covering 465,966 inland lakes worldwide from 2000–2024.
- Developed a large-scale pipeline to acquire and process hundreds of terabytes of Terra MODIS imagery, including parallel downloading and polygon-based ROI extraction.
- Analyzed global spatiotemporal trends using GLCD250-MOD and evaluated the response and interaction mechanisms of lake ecosystems with conventional environmental variables under climate change.

Undergraduate Research Assistant, Water Environment Management Lab Sep. 2023 – Aug. 2025

- Conducted a long-term spatiotemporal trend analysis using the Mann-Kendall test.
- Constructed a Lake Erie Chlorophyll-*a* dataset using open-source data from in-situ and satellites (ENVISAT MERIS, Aqua/Terra MODIS, Sentinel-3A/B OLCI, S-NPP/NOAA-20 VIIRS).
- Projected mid-to long-term algal blooms using Shared Socioeconomic Pathways (SSP).
- Developed a prediction model for cyanobacteria cell counts using Bayesian networks.
- Created a web-specific automatic crawling code.

TEACHING AND MENTORING EXPERIENCE

Engineering Statistics
Teaching Assistant Mar. 2026 – Jun. 2026
Answered course-related questions, created examples and homework programming codes.

Graduation Thesis Class for Undergraduate Students
Academic Mentor 2026
Guided three undergraduates to successfully complete their bachelor's thesis.
- Thesis 2026: Correlation Analysis between Drought Indices and Algal Bloom Indices for Global Lakes

HONORS & AWARDS

Graduate

Outstanding Oral Presentation Award	Feb. 2026
Graduate Student Research Competency Enhancement Program, Spring	Apr. 2026 – Jul. 2026
Instruction Assistance Scholarship	Mar. 2026 – Jun. 2026
The Combined BS/MS Degree Scholarship (Full tuition)	Sep. 2025 – Feb. 2027

Undergraduate

Outstanding Conference Poster Award	Feb. 2025
Outstanding Graduate Thesis Award	Feb. 2025
Grand Prize, Innovative Talent Cultivation Project for Digital Water Industry	Dec. 2024
Student Future Planning Semester: Research Internship	Jul. 2024 – Aug. 2024
Innovative Talent Cultivation Project for Digital Water Industry	Sep. 2023 – Dec. 2024
Student Future Planning Semester: Global Self-Directed Research	Sep. 2023 – Dec. 2023

PUBLICATIONS

[1] **Yang, S.**[1], Lee, H.[1], Lee, G., Gu, T., Park, T., Park, J., Shin, J., Kim, T., Cha, Y. (2026). GLCD250-MOD: Global Daily 250 m Chlorophyll-*a* Dataset for Inland Lakes based on MODIS, *Earth System Science Data (In review)*.

[2] Park, T., Jung, J., Park, J., **Yang, S.**, Moon, J., & Cha, Y. (2026). Structurally Constrained Hierarchical Learning for Improving River Health Assessment using Tabular Transformers. *Water Research,* 126872.

[3] Kim, T., Lee, H., **Yang, S.**, Lee, G., Shin, J., & Cha, Y. (2025). Long-term Spatiotemporal Variability and Regime Classification of Chlorophyll-*a* Concentrations in Lake Erie Using Satellite Products. *Harmful Algae*, 102896.

CONFERENCES

[1] **Yang, S.**, Lee, H., Gu, T., & Cha, Y. Development of a Long-Term, High-Spatiotemporal-Resolution Chlorophyll-*a* Dataset for Global Lakes Using Satellite Observations. Korean Society of Environmental Engineers Winter Conference, 2026.

[2] **Yang, S.**, Lee, H., Kim, T., & Cha, Y. Long-term Trend Analysis of Algal Blooms in Global Lakes. International Water Association World Water Congress & Exhibition, 2026.

[3] Lee, H., **Yang, S.**, Kim, T., & Cha, Y. Construction of a Long-term High Spatiotemporal Resolution Chlorophyll-*a* Dataset for Global Lakes. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Spring Conference, 2026.

[4] **Yang, S.**, Lee, H., Park, T., Gu, T., Moon, J., Park, S., & Cha, Y. Advancement of Aquatic Ecosystem Model Development and Integration Strategies within a Digital Twin Korea-Water. Korean Society of Limnology Winter Conference, 2026.

[5] Kim, T., Lee, H., **Yang, S.**, & Cha, Y. Does drought intensify algal blooms in global lakes? American Geophysical Union Winter Conference, 2025.

[6] Lee, H., **Yang, S.**, Jeong, J., Hwang, G., Moon, J., & Cha, Y. Development of a Machine Learning-Based Geographically Weighted Model for Predicting a Integrated Assessment Index of River Health. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Fall Conference, 2025.

[7] **Yang, S.**, Park, T., Park, J., Gu, T., & Cha, Y. Habitat Suitability Assessment and Derivation of Optimal

Environmental Conditions in the Geum and Nakdong River Basins Using a Data-Driven HSI Model. Korean Society of Limnology Summer Conference, 2025.

[8] **Yang, S.**, Gu, T., Park, T., Lee, H., & Cha, Y. Development of Mid-to Long-term Forecasting Model of Harmful Algal Blooms Based on Probabilistic Predictions Using a Bayesian Network. Korean Society of Limnology Winter Conference, 2025.

[9] Kim, T., Lee, G., Lee, H., **Yang, S.**, Shin, J., & Cha, Y. Characterizing Long-term Spatiotemporal Variability and Regime Heterogeneity of Chlorophyll-a in Lake Erie Using Multiple Satellite-based Products. American Geophysical Union Winter Conference, 2024.

[10] **Yang, S.**, Lee, H., Kim, T., Shin, J., & Cha, Y. Spatiotemporal long-term pattern analysis and regime classification in Lake Erie using an optimal satellite product. Korean Society of Environmental Engineers Winter Conference, 2024.

[11] **Yang, S.**, Lee, H., Lee, G., & Cha, Y. Application of Explainable Deep Learning Model for River Eutrophication Assessment. Applied Artificial Intelligence Conference, 2024.

[12] **Yang, S.**, Jeong, B., & Cha, Y. Analysis of the relationship between the exogenous factor to the Harmful Algae Blooms in the Sejong Weir based on PLS-SEM. Applied Artificial Intelligence Conference, 2023.

PROJECTS

[1] Green Resource Engineering and Advanced Transformation (GREAT) Center for Waste-to-Wealth, National Research Foundation, Graduate Researcher, Jul. 2026 –Present

[2] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (II), National Institute of Environmental Research (NIER), Graduate Researcher, Jun. 2026 – Mar. 2027.

[3] Digital Twin Korea-Water Development Project (5/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (II), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2026 – Jan. 2027.

[4] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (I), National Institute of Environmental Research (NIER), Graduate Researcher, May 2025 – Feb. 2026.

[5] Digital Twin Korea-Water Development Project (4/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (I), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2025 – Dec. 2025.

[6] Development and verification of digital twin-based water quality prediction, National Information Society Agency (NIA), Undergraduate Research Assistant, May. 2024 – Dec. 2024.

[7] Development and Pilot Application of a Digital Twin Aquatic Ecosystem Model (II), National Institute of Environmental Research (NIER), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.

[8] Development of a modularized AI-based model for the multiscale analysis of multimodal environmental data, Ministry of Science and ICT (MSIT), Undergraduate Research Assistant, Mar. 2024 – Feb. 2025.

[9] Water Policy Outlook Based on Integrated Analysis of Water, Environment, and Economy(I), Korea Environment Institute (KEI), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.

SKILLS

Software

Python, R, ArcGIS, Google Earth Engine, Netica, & Crawling.

Modeling Techniques

Bayesian statistics, Bayesian network, Structural equation modeling, Deep learning, & Machine learning.

Languages

Korean (native), English (advanced), Spanish (elementary), & Mandarin (elementary).

ACTIVITIES & INTERESTS

Falcons, University of Seoul baseball club

Club Member Sep. 2020 – Present

Played baseball as a second baseman and pitcher.

Other Interests

Baseball, Flute, and Trekking

MILITARY SERVICE

Republic of Korea Army Jan. 2021 – Jul. 2022

Completed military service as a drill instructor for basic military training.

REFERENCES

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